



# Method of Experimentally Obtaining a Complete Stress-Strain Curve of Masonry Structures

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**Abstract.** The paper describes the methodology of constructing the complete deformation diagram (stress-strain curve) of masonry using hydraulic presses without a tracking system (displacement control) and fast feedback. The technique of equilibrium tests is used, when during deformation of the tested specimen an additional element of increased stiffness is introduced, which allows redistributing the forces when realizing the downward branch of the deformation diagram of the masonry specimen. The paper presents a methodology for selecting the required stiffness taking into account the convexity of the deformation diagram on the downward section. This allows selecting optimal parameters of the loading system, which is especially important in case of a large nomenclature of stone products and variation of their deformation characteristics. The method is tested on thickened silicate bricks with hollowness. According to the results of experimental studies with the developed installation, complete deformation diagrams of masonry under uniaxial compression are successfully obtained.

**Keywords:** Masonry · Stress-Strain Curve · Deformation Diagram · Softening

## 1 Introduction

The construction of full deformation diagrams (stress-strain curve) of structural materials is an important task to ensure the reliability of building structures. The use of physically nonlinear material models allows to establish strength reserves, as well as to determine the ability of a structure to dissipate energy under dynamic loads.

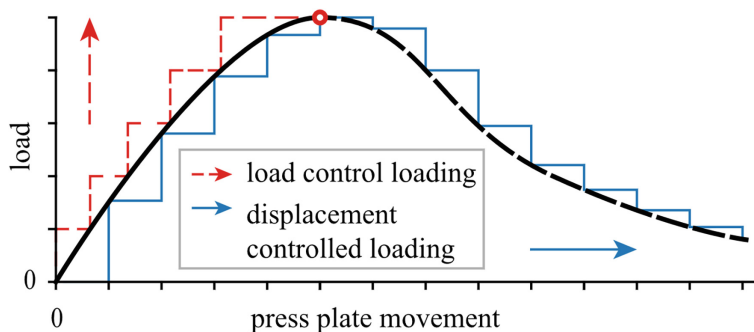
Orthotropic flow theory models realizing the plane stress state [1] are known among the continuum models for masonry as well as failure under load action from the masonry plane [2]. In the set of deformation models, we can distinguish the orthotropic model of masonry [3], as well as isotropic deformation models of concrete [4, 5], which can be successfully applied to masonry under certain types of stress states [6]. For all these models, one of the important parameters is the dependence of stresses on strains in uniaxial compression and tension.

The paper [7] shows the importance of obtaining the plastic characteristics of masonry to substantiate the ability of the material to dissipate energy under seismic

loads, the author obtains the corresponding dependencies numerically on the basis of the discrete finite element model of masonry for plane stress state developed by him [8].

In view of the large range of stone products and mortars used for the construction of masonry walls, there is a problem of normalization of the full deformation diagram of masonry. This problem is reflected both in Russian and foreign norms. The deformation diagram is described only on the ascending section. Also at the moment there is no developed strategy for experimental obtaining of the complete deformation diagram of masonry, which also complicates the use of modern physically nonlinear material models.

One of the problems of experimental studies of masonry structures is the size of a representative volume of masonry whose macromechanical behavior can be used with the required reliability for continuum plasticity models. This imposes limitations on the working areas of the testing equipment and the required levels of failure loads. In the known works on obtaining the full deformation diagram of masonry [9, 10], tests were performed on relatively small cube or column specimens, using testing machines with a tracking system (displacement control) (Fig. 1) and fast feedback. The use of specialized testing machines is a problem to the spread of the experimental method of constructing the descending section of the masonry deformation diagram.



**Fig. 1.** Load-displacement relationship for different test strategies.

Tests with classical hydraulic presses are carried out with loading step control, i.e. the specimen is loaded in equal portions with continuously increasing load (Fig. 1). This leads to the fact that when the ultimate strength of the specimen is reached, the failure is dynamic, due to elastic energy stored in the specimen, which does not allow obtaining the required deformation diagram on the downward section under quasi-static loading.

In order to obtain a complete deformation diagram, there are known methods of equilibrium tests, set forth, in particular, in GOST 29167-2021 for determining the characteristics of crack resistance (fracture toughness) of concrete under static loading. The paper describes the use of high stiffness testing machines or standard testing machines equipped with an additional redistributing device. It is the principle of the redistributing device that is described in this paper for determining the complete deformation diagram of masonry.

In [11], a review of known designs of redistributing devices for concrete testing is given, and the design of an experimental installation in the form of a duralumin alloy pipe, which also acts as a strain gauge force sensor when selecting the gap between the loading plate and the pipe, is proposed. In work [12] as elastic inserts, which perceive forces on the downward section of deformation, calibrated struts are used, on the lower ends of which a stop thread is provided, allowing to regulate the gap between them and the loading plate and include them in operation at load levels of 0.5...0.6 of the destructive ones. In [13], an installation for calculating the modulus of slump was developed, which allows the specimen to be loaded with both tensile and compressive loads using a reverser. The use of this installation is limited to low-grade concrete, because when testing high-strength concrete, more energy is accumulated in the elastic elements of the installation than is required to fracture the concrete. In [14], this principle is used to determine the dynamic modulus of unstrengthening, where it is noted that when the stiffness of the test setup exceeds the unstrengthening modulus, it is possible to obtain a descending section of the deformation diagram in the quasi-static regime up to complete failure of the specimen.

In the case of masonry specimens, the disadvantages of the above-described installations are the required specimen size. Thus, when using a closed tube structure, the specimen size is limited by the size of the tube itself, and the use of compressed steel struts is ineffective under high compressive loads due to loss of stability.

The paper describes the method of selecting the parameters of the experimental installation for determining the full deformation diagram of masonry under uniaxial compression using a universal testing machine. In this case, the principle of operation is similar to that described earlier and allows obtaining the descending part of the deformation diagram due to the redistribution of forces between the specimen and the additionally introduced steel element.

## 2 Methods and Materials

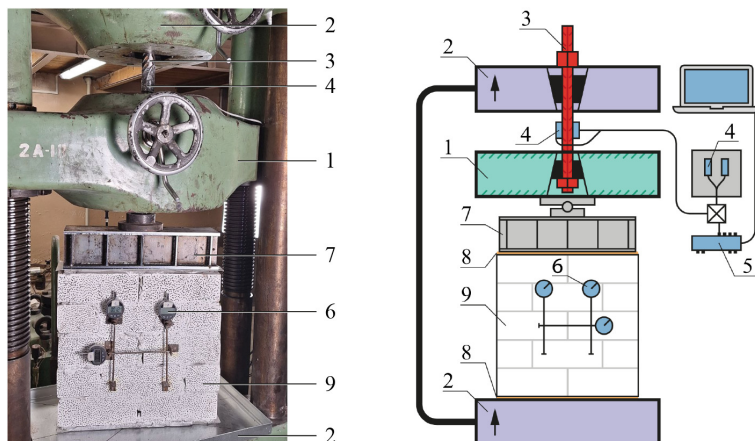
The full deformation diagram was constructed for a masonry specimen of  $510 \times 510 \times 120$  mm in size, manufactured according to GOST 32047-2012. The thickened silicate brick with a void space of 9% of M150 grade according to GOST 379-2015 and mortar with an average compressive strength of 28.4 MPa were used. Characteristics of masonry materials are given in Table 1.

**Table 1.** Strength characteristics of brick and mortar.

| Material | Test method                   | Compressive tensile strength, MPa | Bending tensile strength, MPa |
|----------|-------------------------------|-----------------------------------|-------------------------------|
| Brick    | GOST R 58527-2019             | 15.5 (11%)                        | 3.1 (17%)                     |
| Mortar   | GOST 5802-86<br>GOST 310.4-81 | 28.4 (2%)                         | 7.8 (16%)                     |

The value of the variation coefficient is given in parentheses.

The tests were carried out on the universal testing machine CD100. The scheme of the experimental installation is shown in Fig. 2.



**Fig. 2.** Scheme of the experimental installation: 1 – elevating beam, 2 – fixed beams, 3 – reinforcing bar, 4 – active (on reinforcement) and passive strain gauges, 5 – strain-gauge station, 6 – clock type sensors, 7 – load-bearing girder, 8 – teflon (PTFE) gaskets, 9 – masonry specimen.

A masonry specimen is installed in the compression zone of the testing machine, the relative deformations of the specimen are recorded by six hour-type sensors installed on both sides of the specimen according to GOST 32047-2012. The compressive load is transferred from the fixed crosshead through a hinge to the steel girder and then to the specimen through a teflon (PTFE) gasket. In [15], a good effect of such spacers was shown to minimize the effect of strain constriction in the restrained zones of the specimen and to obtain a uniaxial stress state throughout its volume.

A reinforcing bar of A500C class according to GOST 34028-2016 with nominal diameter of 36 mm with threads and nuts cut on both sides is installed in the tensile zone of the testing machine. The presence of wedges of the testing machine is insufficient to fix the bar, which will be indicated in the results section. The rebar parameters (diameter, strength class and length) were selected based on the experimental methodology described below.

Before the load reaches a value of 0.8...0.9 of the destructive load, the upper nut of the reinforcing bar is released and the masonry specimen is deformed independently. This makes it possible to establish the initial elasticity characteristics (elastic modulus and Poisson's ratio) exactly as required by the normative documents. Then the nut is tightened and compression of the specimen is accompanied by tension of the reinforcing bar. When the masonry strength limit is reached, its stiffness decreases and part of the load is redistributed to the reinforcing bar, the acting forces in which are recorded by means of strain gauges installed on it, the bar itself must be pre-calibrated.

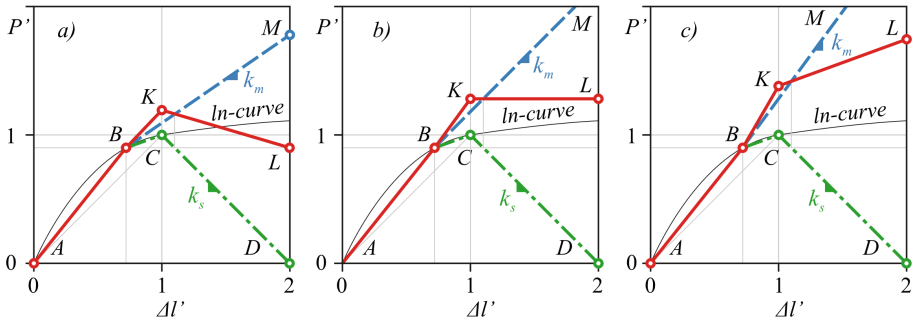
In order to determine the required theoretical stiffness of the reinforcing bar, the complete deformation diagram of the masonry specimen must be given. The diagram can be approximated by the broken  $ABCD$  in Fig. 3. As a basic dependence on the ascending section  $ABC$  (point  $B$  defines the moment of the reinforcement bar inclusion in the work, 0.8...0.9 of the strength of the masonry specimen) can be used the logarithmic dependence presented in SP 15.13330-2020, which in the coordinates of the force-displacement dependence  $P-\Delta l$  (Fig. 3) will take the form:

$$\frac{\Delta l}{l_0} = -\frac{1,1}{\alpha} \ln\left(1 - \frac{P}{1,1R_uA}\right), \quad (1)$$

In [9] it is shown that for masonry on cement mortars without the addition of plasticizing additives, for example, lime, the limiting relative strain is two times higher than the relative strain at reaching the tensile strength, i.e., the unstrengthening modulus can be equal to the secant strain modulus at reaching the tensile strength.

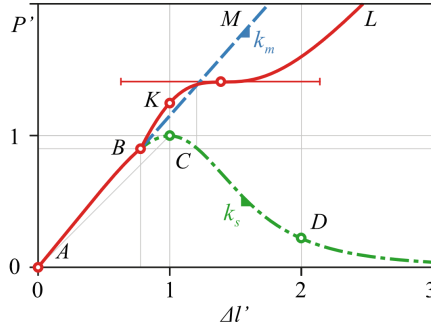
### 3 Results and Discussions

For a stable deformation process, the load-displacement dependence of the press plate (broken  $ABKL$  in Fig. 3) should increase throughout the deformation process, up to the complete failure of the specimen. Then the required theoretical stiffness of the bar  $k_m$  should not be less than the stiffness of the masonry specimen  $k_s$  on the downward section of the deformation diagram (Fig. 3b, c):



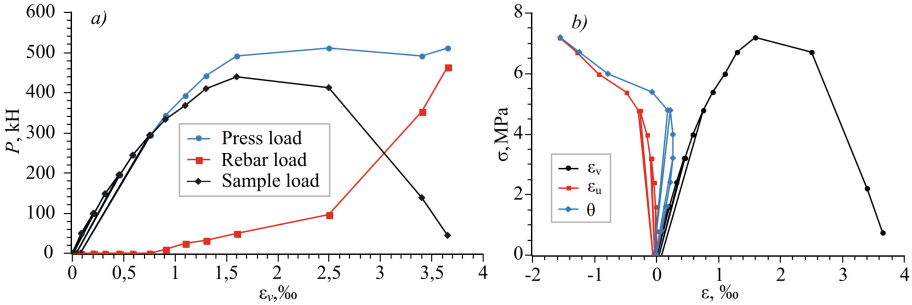
**Fig. 3.** Dependences of press plate displacement on  $ABKL$  load at different reinforcement bar stiffness: a – insufficient stiffness; b – ultimate stiffness; c – optimal stiffness.

It is important to note that the assumption of  $k_m = k_s$  is incorrect due to the upward convexity of the typical diagram of masonry deformation in compression. In general, the first derivative of the graph of the load-displacement relationship of the press plate (curve  $ABKL$  in Fig. 4) should be positive, in which case the value of the required stiffness will be  $k_m \geq 1,1 k_s$ .



**Fig. 4.** Dependences of press plate displacement on load  $ABKL$  at curvilinear diagram  $ABCD$  of specimen deformation.

An experimental study was carried out on the installation shown in Fig. 2. The reinforcing bar was put into operation at stresses of about 0.8 of the ultimate ones. The forces in the specimen were obtained by subtracting the force in the reinforcement bar from the total force produced by the testing machine. As a result, the desired full deformation diagrams of the masonry specimen were constructed and are shown in Fig. 5. Since the main cracks open on the downward section of the diagram, it is difficult to determine the strains using clock-type sensors, so it is either necessary to record the movements of the press plate or to use the calibrated dependencies for the reinforcing bar. This is possible because after sampling the compression strains, the strains of the reinforcement and the specimen are combined.



**Fig. 5.** Dependence of forces in the elements of the experimental installation on relative deformations of the masonry specimen (a) and obtained deformation diagrams (b)

## 4 Conclusions

As a result of the conducted research, a methodology for constructing a complete deformation diagram of masonry has been developed and tested. The experimental installation allows equilibrium tests to be performed using universal testing machines not equipped

with a tracking system (displacement control) and fast feedback. The analytical method of steel bar parameters selection allows determining the required stiffness for different types of masonry units based on the assumed deformation diagram or on the basis of the first non-equilibrium test to determine the primary elastic characteristics and the values of ultimate deformations when the material strength limit is reached.

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